A FINITE GRADED MODULE OVER A GRADED POLYNOMIAL ALGEBRA THAT IS FLAT OVER A LOCAL RING IS FINITELY PRESENTED

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Allowed standard prerequisites (editorial).

- A finite flat module over a local ring is free. Algebra, Lemma \ref{algebra-lemma-finite-flat-local}, cited at proof lines 5884 and 5899. Makes each graded piece of the module, and then each graded piece of the kernel, a finite free module.
- Flatness passes to the kernel of a surjection of flat modules. Algebra, Lemma \ref{algebra-lemma-flat-ses}, cited at proof lines 5895-5897. Shows that each graded piece K_d of the kernel is flat over R.
- Nakayama's lemma. Algebra, Lemma \ref{algebra-lemma-NAK}, cited at proof line 5907. Turns a generating set of the kernel after tensoring with the residue field into a generating set over the ring.

Source reference. Original lemma 053C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6600–6600. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00NZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 877–877. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00HM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 648–648. Navigation pointer to a statement the

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Source reference. Original lemma 00DV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 512–512. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let $S = R[x_1, \ldots, x_n]$ be a graded polynomial algebra over R, i.e., $\deg(x_i) > 0$ but not necessarily equal to 1. Let M be a graded S-module. Assume*

(1) *R is a local ring,*
(2) *M is a finite S-module, and*
(3) *M is flat over R.*

Then M is finitely presented as an S-module.

Proof. Let $M = \bigoplus M_d$ be the grading on M. Pick homogeneous generators $m_1, \ldots, m_r \in M$ of M. Say $\deg(m_i) = d_i \in \mathbf{Z}$. This gives us a presentation

$$0 \to K \to \bigoplus\nolimits_{i=1,\ldots,r} S(-d_i) \to M \to 0$$

which in each degree d leads to the short exact sequence

$$0 \to K_d \to \bigoplus\nolimits_{i=1,\ldots,r} S_{d-d_i} \to M_d \to 0.$$

By assumption each M_d is a finite flat R-module. By Algebra, Lemma 00NZ this implies each M_d is a finite free R-module. Hence we see each K_d is a finite R-module. Also each K_d is flat over R by Algebra, Lemma 00HM. Hence we conclude that each K_d is finite free by Algebra, Lemma 00NZ again.

Let $\mathfrak{m}$ be the maximal ideal of R. By the flatness of M over R the short exact sequences above remain short exact after tensoring with $\kappa = \kappa(\mathfrak{m})$. As the ring $S \otimes_R \kappa$ is Noetherian we see that there exist homogeneous elements $k_1, \ldots, k_t \in K$ such that the images $\overline{k}_j$ generate $K \otimes_R \kappa$ over $S \otimes_R \kappa$. Say $\deg(k_j) = e_j$. Thus for any d the map

$$\bigoplus\nolimits_{j=1,\ldots,t} S_{d-e_j} \longrightarrow K_d$$

becomes surjective after tensoring with κ. By Nakayama's lemma (Algebra, Lemma 00DV) this implies the map is surjective over R. Hence K is generated by $k_1, \ldots, k_t$ over S and we win. □